

Artificial Neural Networks for Snow Avalanche Forecasting in Indian Himalaya

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ABSTRACT: Snow avalanches pose serious threat to Indian troops deployed in snow-bound areas of western Himalaya during winter months. The most viable way to mitigate avalanche threat in these areas is to precisely predict the time and place of their occurrences. Since, the factors involved in the formation of an avalanche are too many and underlying physical processes are quite complex, no prediction model can completely imitate the thought process and analysis methods of an expert forecaster. However, with the recent developments in the field of computational capabilities and Artificial Intelligence, improved models are coming up and proving to be reliable assistant to avalanche forecasters. In this direction, an attempt has been made to exploit the technique of Artificial Neural Networks (ANN). The base model is inspired by the popular non-parametric k-Nearest Neighbours method, where the two-parameter output is processed through ANN for classification. The model has been tested for an avalanche prone sector of Kashmir region and results are quite satisfactory. The paper discusses in detail the overall methodology adopted in the development of present model, and test results observed. Further possible use of Artificial Intelligence techniques in avalanche forecasting has also been discussed.

1 Introduction

The Indian army troops deployed in high altitude snow bound regions of Western Himalaya remain under grave danger of snow avalanches, also known as 'white death', during winter months. Snow & Avalanche Study Establishment (SASE) is committed to ensure the safe mobility of Indian troops in these areas, and has been untiringly working towards the mitigation of this danger through both active and passive methods. Whereas, the active method of installation of control structures offers permanent solutions, it is very difficult to implement and often economically unviable. The other active method of controlled release of avalanches by use of explosives also has practical limitations and thus is practiced only on select suitable locations. The most viable and comprehensive solution to this problem in such a vast and difficult terrain coupled with harsh weather is avalanche forecasting, a passive method. The idea is to avoid the danger when it is likely to hit. Thus the very aim of avalanche forecasting is to precisely predict the time and site of avalanche occurrence and thus avoid it.

Avalanche Forecasting is an expert task and involve collection and analysis of a variety of snow and meteorological data from a number of representative observatories in the area. The governing parameters are too many and underlying physical and mechanical processes are highly complex and not yet fully understood. Added to it is the high variability in the characteristics of snow cover across the slopes. Thus, the experts by virtue of their field exposure and knowledge about local terrain are able to predict much better than any physical or numerical model. However, with the recent developments in computational capabilities and Artificial Intelligence techniques, improved and intelligent models are coming up and proving as reliable tools in the hands of avalanche forecasters.

For avalanche forecasting, Nearest Neighbors (NN) Method has been very popular among avalanche forecasters since its introduction by Buser (1983). It is more popular as a data-mining or information gathering tool than as a classification tool, for its outstanding ability to provide the forecasters with all important information related to previous similar occurrences. Thus, the focus of attention so far has been on the qualitative information available from nearest past days brought out by the model and no major efforts were made to use the method as classification tool. A few workers (Gassner *et. al.*, 2000; Purves *et. al.*, 2003) determined the avalanche occurrence probability as the ratio of number of avalanche days to the total number of nearest days and then used a threshold probability value for classification but success rate was limited. It may be noted here that, since k-nearest neighbors have varying distances from the data point under consideration, they cannot be considered equally probable for any further statistical inferences. Hence, as per the concept of probability, the simple ratio of

avalanche days to the total number of nearest days does not represent the true probability of avalanche occurrence. To address this issue to some extent, McCollister *et. al.* (2002) introduced the concepts of weighting more similar days more heavily by the inverse function of distance. Singh *et. al.* (2005) weighted the nearest days with a function of its rank to assign more weight to more near neighbors.

Taking inspiration from the above, we have now introduced a more realistic approach for probability determination from nearest neighbors where we employ the relative positions of nearest neighbors from the data point under consideration. With the modification as mentioned above, probability of occurrence of avalanche is not complimentary to probability of non-occurrence of avalanche. Thus, probability of non-occurrence of avalanche may also be taken as an indicator for classification. For final classification, the technique of Artificial Neural Networks (ANN) has been employed which takes into account respective probabilities for both occurrence of avalanche and non-occurrence of avalanche as brought out by the NN model. The method has been tested for Chwokibal-Tangdhar sector of Kashmir region in Indian Western Himalaya and results have been discussed in subsequent sections.

2 Study Area

Chowkibal-Tangdhar sector (CT Sector) in Kashmir region of Indian Western Himalaya (Figure 1) was chosen for assessing the overall proficiency of the proposed model. For this purpose, the snow-meteorological data from Stage-II observatory (altitude 2650m) of this sector has been used to represent the sector. A total of 17 avalanche sites (CT-1 to CT-17) affect the section of CT sector considered for the study.

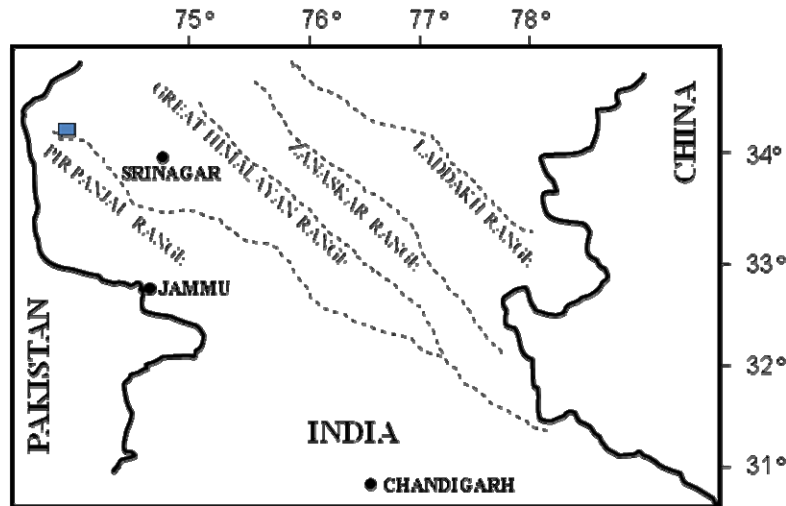


Figure 1: Chowkibal-Tangdhar sector (Jammu & Kashmir) shown as shaded box.

3 Model Features

3.1 Distance Metric and Input Parameters

Standard Euclidian distance metric as described below has been used for calculating distance (d) between data points –

$$d = \sqrt{\sum_{i=1}^n w_i \Delta x_i^2} \quad (1)$$

Where,

n = Total number of input parameters

w_i = weight assigned to i^{th} parameter

Δx_i = Difference of normalized values of i^{th} parameter

The normalization of parametric values is carried out from their respective maximum and minimum values so as to bring all the parameters on a common scale of 0 to1. The parameters used and their respective weights are given in Table 1.

Table 1. Nearest neighbours model input parameters, their units, observation periods/time (as on any day-t), and weights assigned.

No	Parameter (P_i)	Unit	Observation period/time	Weight
1.	Air Temperature	°C	0830h(t)	1
2.	Snow Surface Temperature	°C	0830h(t)	1
3.	Air Temperature Change	°C	0830h(t) – 0830h(t-1)	1
4.	New Snow in 24 hours	cm	0830h(t-1) to 0830h(t)	3
5.	New Snow in 48 hours	cm	0830h(t-2) to 0830h(t)	2
6.	New Snow in 72 hours	cm	0830h(t-3) to 0830h(t)	1
7.	Snowpack Depth	cm	0830h(t)	1
8.	Snowpack Water Equivalent	mm	0830h(t)	2
9.	Wind Speed	m/s	0830h(t)	3
10.	Free Ram Penetration	cm	0830h (t)	1

3.2 Probability determination

For classification through NN model, the traditional approach has been to find the probability of avalanche occurrence (P_{av}), given k-nearest neighbours, as –

$$P_{av} = \frac{\text{No. of avalanche days out of k nearest neighbours}}{k} \quad (2)$$

Based upon the output from training data set, a threshold probability is then decided by the experts and classification is carried out for test data.

Now, in the present approach we have taken it to a step further and incorporated the relative distance of neighbours from the datapoint under consideration to give more weight to the nearest points. For this purpose the distance of farthest point in the database has been taken as reference distance. Also, we determine both probability of avalanche occurrence (P_{av}) and probability of non-occurrence (P_{nav}) for classification, as they are not complimentary once the weights have been assigned to the neighbours as per their relative positions. Mathematically it can be represented as –

$$P_{av} = \frac{\sum_{i=1,k}^{aval} (1 - d_i/d_{max})}{k} \quad (3)$$

Where,

k = total number of neighbours selected

d_i = absolute distance of i^{th} neighbor

d_{max} = absolute distance of farthest neighbor, and summation is only over the avalanche-day neighbors i.e.

$d_i = d_{max}$ for non-avalanche neighbors.

Similarly,

$$P_{nav} = \frac{\sum_{i=1,k}^{non-aval} (1 - d_i/d_{max})}{k} \quad (4)$$

Where, summation is only over non-avalanche neighbours i.e. $d_i = d_{max}$ for avalanche neighbors.

Thus, a set of combinations of both P_{av} and P_{nav} should be able to offer better classification criteria.

3.3 Database

The past data of input parameters observed at Stage-II observatory and as described in Table 1 has been archived in a database with respective dates of observation. Also archived with each date is the associated avalanche information (Occurred/ Not occurred, size, distance traveled, aspect, natural/artificial etc.). There are also some cases, where avalanche feedback was not precisely known due to certain practical limitations, but

according to the field experts avalanches would have triggered on such days. These days have been specially marked and carry 50% weight for determination of P_{av} and P_{nav} , if selected as nearest neighbor.

3.4 Artificial Neural Networks for Classification

Artificial neural Network (ANN) was designed as described in Figure 2 to classify the test data in either of the two classes – ‘Avalanche Day’ and ‘No Avalanche Day’ based on two input parameters – P_{av} and P_{nav} .

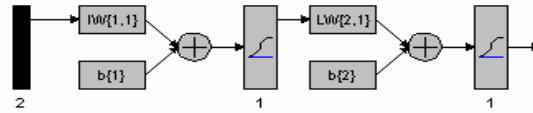


Figure 2: Artificial Neural Networks designed for classification

The Log-Sigmoid Function as described in Figure 3 has been employed as transfer function at hidden layer level as well as output level.

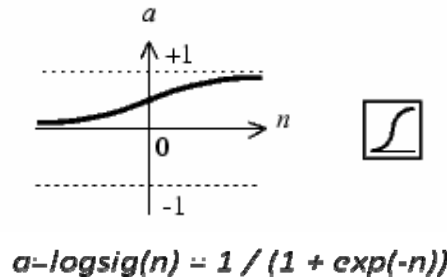


Figure 3: Log-sigmoid transfer function

For learning of ANN, P_{av} and P_{nav} values for the winters from 2002-03 to 2005-06 (Dec to Mar only) were first calculated through NN model as described above. The values thus obtained, as shown in Figure 4, were used to train the ANN in supervised learning mode, for dichotomous target - 1 for ‘Avalanche Day’ and 0 for ‘No-avalanche Day’. The various parameters thus obtained are presented in Table 2.

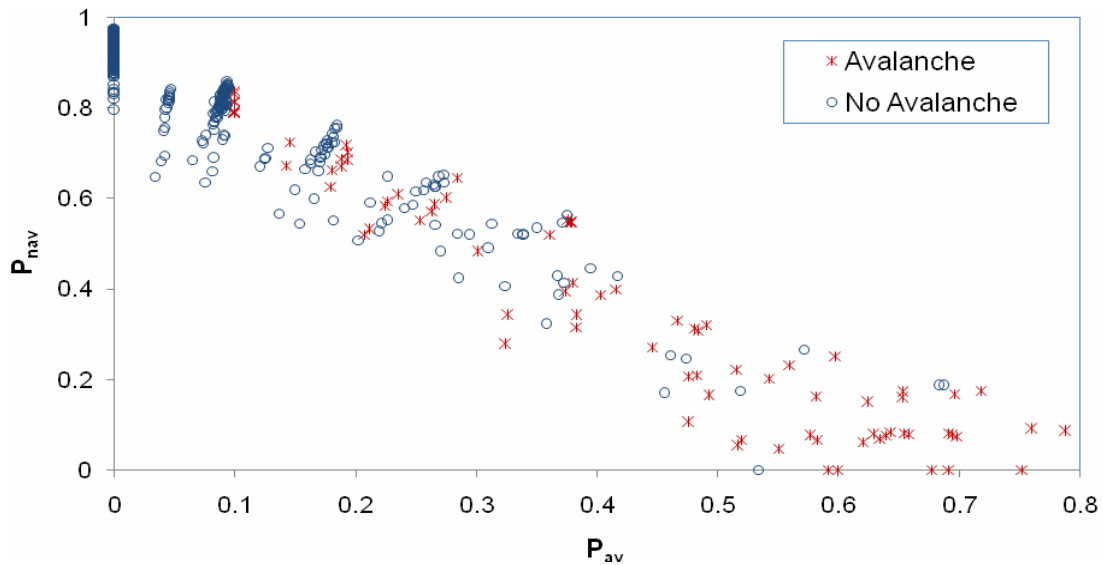


Figure 4: Set of P_{av} and P_{nav} corresponding to CT sector used for training the Artificial Neural Network (2002-03 to 2005-06 and from Dec to Mar only each winter)

Table 2: ANN parameter obtained after learning.

Parameter	Description	Value
IW{1,1}	Weight to layer-1 from input-1	-1.2455, 0.32764
LW{2,1}	Weight to Layer-2 from layer-1	-13547.6461
B{1}	Bias to layer-1	-7.6872
B{2}	Bias to layer-2	4.7132

4 Test Results and Discussion

The ANN output for test data of CT sector during 2006-07 winter (01 Dec, 2006 to 31 Mar, 2007) in Generalization mode is shown in Figure 5 along with the actual observations. Since the ANN output is on a continuous scale from 0 to 1, a threshold value is required for crisp classification and also to test the model performance in quantitative terms.

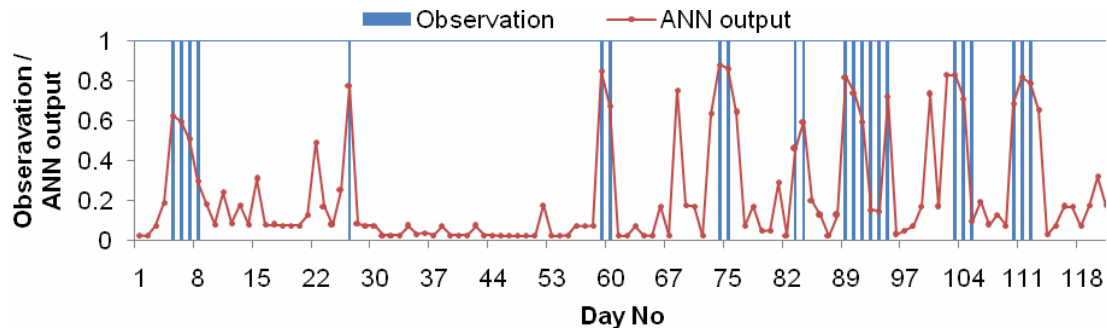


Figure 5: ANN output for test data ('Day No' starting from 01 Dec, 2006).

For analyzing the model performance following two accuracy measures have been considered –

- (i) Probability of Detection (POD)
- (ii) False Alarm Rate (FAR)

Probability of Detection (POD) may be defined as the probability that the event was forecasted when it occurred. Similarly the False Alarm Rate may be defined as the probability that event did not occurred when it was forecasted. Thus with the help of these two accuracy measures the performance of overall methodology can be analysed quite reliably in quantitative terms. Thus, the performance accuracy measures with different threshold ANN output values were calculated and are presented in Figure 6(a).

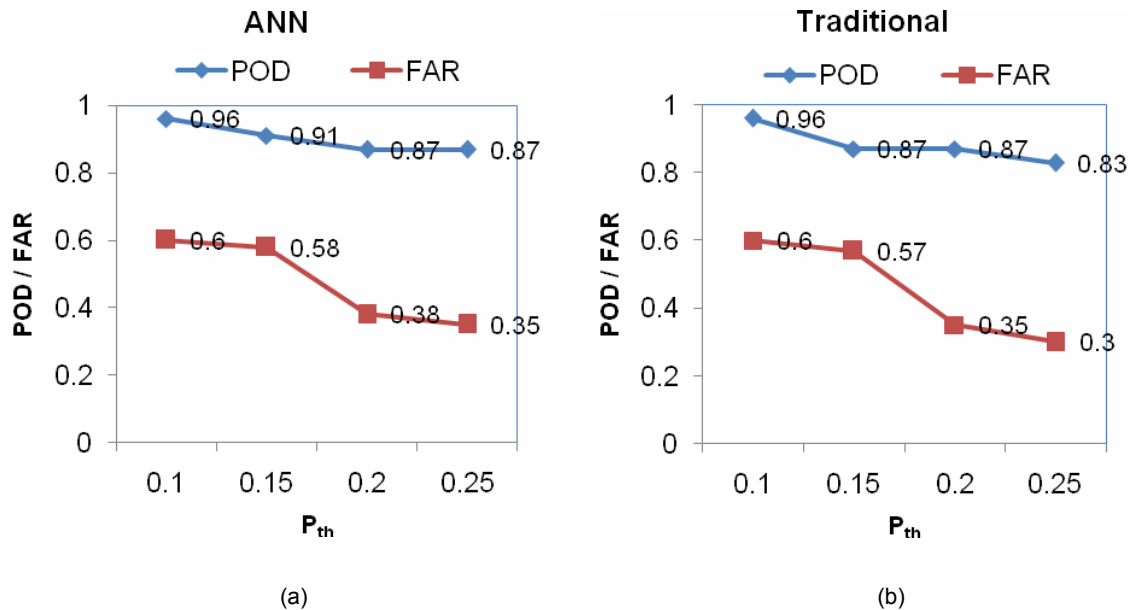


Figure 6: Comparison of accuracy measures – (a) Artificial Neural Network (b) Traditional approach. P_{th} on x-axis is the threshold value for respective approaches and y-axis represent the values of POD and FAR.

At this juncture, it is imperative to see how much improvement has taken place through the present approach over the traditional approach where only P_{av} is used as a decision criterion. Since, the traditional approach uses only P_{av} for decision making, the performance measures with different threshold values of P_{av} on the same dataset are shown in Figure 6(b). On comparing Figure 6(a) and Figure 6(b), it is obvious that at around the same level of FAR of 0.58 the POD through ANN is 0.91 whereas the POD through traditional approach is 0.87.

Our desired performance accuracy measures, for CT sector are $POD \geq 0.90$ with $FAR \leq 0.50$. This means that we should be able to capture at least 90% of the avalanche occurrences in the area and at the same time the total number of predicted 'Avalanche days' should not be more than twice the number of actual 'Avalanche days' observed in the whole season. These figures seems tolerable, keeping in view the fact that in CT sector, from Dec to Mar in each winter around 20% days are avalanche days. Thus, if by keeping the road close for about 40% days of the winter, we are able to capture more than 90% of avalanches, our purpose is satisfactorily met. It is obvious from Figure 6 that with the use of ANN we can satisfactorily meet the desired performance measures with a threshold ANN output value slightly above 0.15. The same seems to be little unfeasible with traditional approach where FAR has already exceeded the desired maximum level of 0.50 at POD level of 0.87.

Besides the above, the other important feature of this approach is the detection of some unique avalanche days which are otherwise not possible to be detected through traditional approach with similar efficiency parameters. As an example, for 02 Mar-2007 the P_{av} , P_{nav} and ANN output are 0.1294, 0.5877 and 0.155. Through traditional approach this day can be classified as avalanche day, only if threshold value is little lesser than 0.1294. But at this threshold value the overall performance will not meet the desired performance as FAR will far exceed the desired value of 0.50 (ref Figure 6).

5 Conclusions

The proposed approach has potential in detecting avalanche days which were otherwise impossible through traditional method of using only the avalanche occurrence probability. The model is simple yet the overall performance meets the desired levels satisfactorily. However, still there is some scope left for further improvement in fixing the decision boundaries. Attempts may be made to use upcoming Artificial Intelligence methods and optimization tools such as Genetic Algorithm etc. to optimize the decision boundaries for better results.

6 References

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